Detecting and Quantifying Surface Waviness on Flowing Liquid Metals

Summary

	Method	Measures	Shown on	Reported performance
1	Laser spots reflected onto a screen	slope, wavenumber, phase speed	liquid Ga (driven waves)	geometry calibrated to <1% [1]
2	Laser-line / laser-sheet triangulation	height profile	Galinstan channels	0.7–2.6% of local depth [2]; 0.15 mm phase-averaged [3]
3	Point probes (ground truth)	height time series	GaInSn, Galinstan, Li, Hg	inductive <1 mm at 20 Hz through an 8 mm wall [4]; ultrasonic ∼0.01 mm [5]; capacitive 0.1 ms [6]
4	High-speed imaging through glass	kinematics, morphology	GaInSn in a glass cell	350 fps; needs oxide control [7]
5	Reflected-pattern deflectometry	full-field slope	water, glycerol, oil only	±4° slope [8]; 1 μm claimed [9]

1 Detection: how waviness has been measured

1.1 Reflected laser spots: the cheapest proven slope measurement on liquid metal

- **How it works.** One 2 mW He–Ne laser is split into up to seven beams aimed at the gallium surface. Their reflections land on a screen, which a gated ICCD camera films at 60 fps with 0.1–1 ms gates [1].
- **What it gives.** A tilted surface moves the reflected spot. The displacement is $\Delta r_0 = 2dka$, where d is the screen distance, k the wavenumber and a the amplitude. The phase difference between neighbouring spots gives k, and from it the phase speed [1].
- **Shown on.** Paddle-driven waves (4–12 Hz) on liquid gallium, with the geometry calibrated in situ to better than 1% [1].
- **Limits.** 60 fps was enough only because the waves had a single known frequency and each run was triggered from the camera's gate signal [1]; freely evolving waves need a faster camera (our inference). The screen must be large enough to catch the deflected spots, which caps the measurable slope, as for other reflection-based slope gauges [8].

1.2 Laser-line triangulation: calibrated height profiles, but it needs some diffuse return

- **How it works.** A laser sheet draws a line across the metal. A camera viewing at an angle sees the line shift as the surface rises and falls, and the shift converts to height by triangulation.
- **Shown on.** Princeton's open Galinstan channels. The first rig calibrated the laser sheet and camera against an electrical contact probe [10]; the larger LMX-U rig slid the sheet along the channel and calibrated against solid blocks of known thickness [11]. Height uncertainty was 0.7–2.6% of the local film height, and 6.9% at worst at single stations [2].
- **Dealing with glare.** Gutiérrez & Aumaître imaged the laser line from two directions (±37°) onto one camera sensor, so a glint cannot blind both views at once. Averaging over more than 100 periods of a single-frequency wave gave 0.15 mm amplitude precision. Their Galinstan sat under about 1% hydrochloric acid to keep it free of oxide [3].
- **Limits.** The camera needs some light scattered back from the surface. Brenneis reports that the oxide's diffuse reflection "benefits or enables" optical sensing: a flat, clean GaInSn surface could not be detected at all, while a surface with small waves gave $\sigma < 0.35$ mm with only 36–56% of points returned [12].

1.3 Point probes: the ground truth behind every quantitative result

- **Inductive sensor (no contact, works through a wall).** Coils outside the vessel induce currents in the metal and sense the level through an 8 mm non-conducting wall. Precision is better than 1 mm at 20 Hz, within 0.5 mm of a laser reference. It averages over the coil footprint and must be calibrated before each use [4]. On the same GaInSn model, a 2.8 Hz ultrasonic level sensor was too slow to follow the surface waves, while a 93 Hz laser scanner and the inductive sensor agreed closely [13].
- **Ultrasonic pulse-echo.** 4 MHz transducers behind a 5 mm PMMA plate send pulses down through a KOH layer to the GaInSn interface, and the echo time gives the local height. They resolved wave amplitudes of about 0.01 mm. A tilted surface reflects the echo away, which caps the measurable amplitude at about 20% of the container radius [5].
- **Resistive electrodes.** Wire electrodes in the Galinstan measure the electrical resistance across the layer, which a formula for wire electrodes converts to thickness; the sensor was calibrated against a ruler. Oxide on the electrodes adds an offset that the calibration removes. The authors state it meets a ∼10 ms response requirement [14].
- **Contact probe.** A needle is stepped vertically into a lithium jet and records how often it touches the metal at each height. Contact frequency versus height gives the wave-height distribution, and its peak marks the mean level [15]. The same programme used a time-of-flight laser sampling at 500 kHz with 0.04 mm uncertainty (conference abstract) [16].
- **Capacitive wire.** An insulated 0.1 mm wire dipped into mercury forms a capacitor with it, so the capacitance tracks

the local height: range 10 μm–2 cm, 0.1 ms response [6]. We found no use in gallium alloys.

1.4 Imaging through a wall: demonstrated only with oxide control and a dark room

- **Through glass.** Jegorovs et al. filmed GaInSn in a 3 mm glass cell at 350 fps, backlit in a darkened room because the metal is "mirror-like". Oxide coated the walls "in a matter of minutes" unless the metal was kept in about 25% HCl, and any metal film wetting the glass is opaque [7].
- **From above, no wall.** Other gallium-alloy studies filmed open surfaces from above: particle velocimetry of the surface at 2000 fps [17], tracking of floating tracers at 120 fps [11], and 60 Hz infrared imaging of oxide particles, which emit far more infrared than clean metal [18].
- **Closest to our flow.** A ∼1 m/s GaInSn channel in air used high-speed video to measure the position and speed of wave patterns, but no wave heights [19].

1.5 Full-field reflected-pattern slope: promising, not yet shown on liquid metal

- **How it works.** The camera sees a known pattern reflected in the liquid. Surface slope distorts the reflection; comparing it with an image of the flat surface gives the slope at every point, which can be integrated to height.
- **Colour-coded screen.** The colour seen in the reflection encodes slope: hue gives its direction and saturation its magnitude. It was demonstrated to ±4° on water. Reflection gives 2° of ray deflection per degree of slope, against about $\frac{1}{4}^\circ$ for refraction, but over a smaller slope range [8].
- **Moon-glade background-oriented schlieren.** The camera and a backlit pattern both sit above the liquid. The authors claim it suits opaque liquids and resolves 1 μm, but tested it only on glycerol, water and silicone oil [9].
- **Not applicable.** Free-surface synthetic schlieren [20] and total-internal-reflection deflectometry [21] need light to pass through the liquid, so they cannot work on an opaque metal (our inference).

2 Quantification: what to compute

Metric	How it is computed and what it shows	Source
Wave-height statistics	Split a height record $\eta(t)$ into waves at each upward crossing of the mean level, then report the mean height $\bar{H}$, the significant height $H_s \approx 1.6\,\bar{H}$ and the maximum $H_{\max}$. On a lithium jet, heights followed a Rayleigh distribution and periods a log-normal one.	[15, 16]
Frequency spectrum	The power spectral density of the level record shows the dominant wave frequencies (peaks at 1.2, 1.8 and 2.1 Hz in a 1/3-scale water model of a casting mould). Nippon Steel compares such peaks with the sloshing modes $f_N = \frac{1}{2\pi}\sqrt{N\pi g/L}$ of a vessel of width L.	[22, 23]
Sampling rate	Sample faster than twice the highest wave frequency: a 2.8 Hz sensor could not follow 2.7 Hz waves.	[4]
Spatial variability	Record the level at several stations and take the standard deviation over time at each, $\sigma_h(x)$. Low values mark standing-wave nodes, and a single sensor placed at a node misses fluctuations elsewhere.	[24]
Asymmetry	$\mathrm{AI} = 100\,(h_L - h_R)/h_{\mathrm{ref}}$, with h_L, h_R the standing-wave heights on each side and $h_{\mathrm{ref}} = 20$ mm. A large spread of AI over time indicates swaying flow.	[25]
Level-band fraction	Percentage of level fluctuation that stays within ±3 mm: 57–74% vs. 92–99% for two nozzle designs in a water model.	[26]
Slope distribution from glints	Cox & Munk derived the distribution of sea-surface slopes from the brightness of sun glitter in photographs; this requires knowing where the light and camera are.	[27]
Slope to height	For laser spots, $a = \Delta r_0/(2dk)$. For a full slope field, integrate by least squares.	[1, 20, 21]
Waviness vs. roughness	A Gaussian profile filter separates the large- and small-scale components of a surface profile.	[28]

3 Oxide control in the cited rigs

- **Gas cover.** Princeton's Galinstan rigs and the KIT sensor tests ran under argon [11, 12, 18]. For pure gallium, Ji et al. replaced the air with nitrogen and argon "without apparent success", but 1 M HCl cleaned a gallium sample [1].
- **Liquid cover.** Others kept the metal under a liquid layer: about 1% HCl [3], about 25% HCl [7] or KOH [5]. The Columbia group skimmed the oxide off and Teflon-coated parts of the setup [14].
- **Why it matters.** With an oxide skin, the reflected signal follows "the slowly evolving membrane" rather than the flow [14]. Ji et al. suggest oxide damps waves [1], and Hiraka et al. suspect it shifted their results between runs [19].

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